Youssef Elhelw

Data Scientist

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Egypt: Cairo

AI student. Passionate about data science, machine learning, and AI engineering, with a strong focus on solving real-world problems. Gained practical experience through projects, internships, and job simulations. Developed solid analytical and problem-solving skills, with a growing interest in fields such as NLP, computer vision, and generative AI.

Education

Bachelor at Computers and Artificial Intelligence at Cairo University, Giza

September 2023 - July 2027

Internships

Data Scientist at Elevvo Pathways

September 2025 - October 2025

- Developed and trained 5 distinct machine learning models, covering a wide range of ML concepts.
- Engineered ML pipelines for regression and recommendation systems, delivering strong predictive performance.

Data Analyst at Minders, Dokki, Egypt

- Acquired practical experience in data analysis through active engagement with Minders, contributing to real-world data-driven tasks and projects.
- Built interactive dashboards to visualize data and support decision-making.

Skills

Programming Skills: Python, SQL, C++, Java, OOP, FastAPI

Math (Linear Algebra, Calculus) & Statistics

Data Science: Data Handling, Data Cleaning, Data Analysis (EDA), Data Visualization, Machine Learning, Deep Learning, Data Preprocessing for ML, Model Deployment

Version Control: Git, GitHub

Soft Skills: Problem-solving, Critical thinking, Communication, Teamwork, Leadership

Projects

Movie Recommendation System • Try it out

December 2025 - February 2026

A movie recommendation web app where users enter a movie they like, and the system recommends 10 movies with the same “vibe.”

- Built an end-to-end Movie Recommendation System that suggests personalized movies using user preferences and content similarity techniques (TF-IDF + Cosine Similarity).
- Designed a hybrid recommendation pipeline combining content-based filtering and feature engineering.
- Developed a scalable similarity-based ranking system to retrieve top-N most relevant movies efficiently.
- Implemented advanced text preprocessing and feature extraction techniques to transform raw movie descriptions into structured numerical representations.
- Improved recommendation quality by applying weighted feature importance (genres, keywords, title, overview) to enhance semantic similarity.
- Built a full pipeline from data cleaning → feature engineering → vectorization → similarity computation → recommendation output.
- Implemented web scraping techniques to collect additional movie data from online sources and fill missing values, enriching the dataset with new movies and improving overall data completeness.

Languages

Arabic - Native

English - Proficient